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Splitter Harness for Vehicles with a Proprietary Interface Port Connector

Abstract

A splitter harness and a method of installing the splitter harness are provided. The splitter harness includes a main harness segment having a first multi-wire harness connector and a second multi-wire harness connector at both ends thereof, an On-Board Diagnostic (OBD) port harness segment having an OBD port connector at an end thereof, a signal splitter, and a power signal provider for providing power to the OBD port connector. The method includes unplugging the vehicle connector cable connectors, connecting the harness connectors, and connecting the power signal provider to a power source.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims priority from U.S. provisional application 63/561,986 filed Mar. 6, 2024, the contents of which are herein incorporated by reference in their entirety.

FIELD

[0002] The present disclosure generally relates to vehicle telematics, and more specifically to a splitter harness for vehicles with a proprietary interface port connector.

BACKGROUND

[0003] A telematics system may gather asset data using a telematics device. The telematics device may be integrated into or located onboard the asset. The asset may be a vehicle (“vehicular asset”) or some other stationary equipment. The telematics device may collect the asset data from the asset through a data connection with the asset. In the case of a vehicular asset, the telematics device gathers the asset data via an interface port, such as an on-board diagnostic port (OBD).

Additionally, the telematics device may gather sensor data pertaining to the asset via sensors on the telematics device. Furthermore, the telematics device may gather location data pertaining to the asset from a location module on the telematics device. When the telematics device is coupled to the asset, the gathered sensor data and location data pertain to the asset. The gathered asset data, sensor data and location data may be received and recorded by a technical infrastructure of the telematics system, such as a telematics server, and used in the provision of fleet management tools, for telematics services, or for further data analysis.

SUMMARY

[0004] In one aspect of the present disclosure there is provided a splitter harness. The splitter harness comprises a main harness segment having a first multi-wire harness connector at a first end thereof, and a second multi-wire harness connector at a second end thereof. The splitter harness also comprises an On-Board Diagnostic (OBD) port harness segment having an OBD port connector at a first end thereof, a signal splitter for splitting a group of signal lines of the main harness segment into a first group of signal lines routed on the main harness segment to the second multi-wire harness connector and a second group of signal lines routed to a second end of the OBD port harness segment, and a power signal provider for providing a power signal from a power source to the OBD port connector. The splitter harness allows connecting a device such as a telematics device having an OBD interface to a vehicle with a proprietary interface port. The splitter harness is built using off-the-shelf parts and is therefore economical. The splitter harness provides power to the telematics device even though the proprietary interface port does not have power signal lines.

[0005] The first multi-wire harness connector may be configured for connecting to a vehicle module multi-wire connector of the vehicle module.

[0006] The second multi-wire harness connector may be configured for connecting to a proprietary interface port multi-wire connector of the proprietary interface port.

[0007] The first multi-wire harness connector may comprise a male plug and the vehicle module multi-wire connector comprises a female receptacle.

[0008] The first multi-wire harness connector may comprise a female receptacle and the vehicle module multi-wire connector comprises a male plug.

[0009] The second multi-wire harness connector may comprise a male plug and the proprietary interface port multi-wire connector comprises a female receptacle.

[0010] The second multi-wire harness connector may comprise a female receptacle and the proprietary interface port multi-wire connector comprises a male plug.

[0011] The power signal provider comprises a power signal line and a fuse tap.

[0012] The signal splitter comprises a splitter connector located on the main harness segment.

[0013] The signal splitter comprises a splitter connector which is part of the first multi-wire harness connector.

[0014] In another aspect of the present disclosure, there is provided a method of installing a splitter harness in a vehicle having a proprietary interface port connected to a vehicle module via a vehicle connector cable. The splitter harness comprises a main harness segment having a first multi-wire harness connector and a second multi-wire harness connector at both ends thereof, an On-Board Diagnostic (OBD) port harness segment having an OBD port connector at an end thereof, a signal splitter for splitting a group of signal lines on the main harness segment into a first group of signal line routed to the second multi-wire harness connector and a second group of signal lines routed to the OBD port harness segment, and a power signal provider for providing power to the OBD port connector. The method comprises unplugging the vehicle connector cable from the proprietary interface port, unplugging the vehicle connector cable from the vehicle module, connecting the first multi-wire harness connector with the vehicle module, connecting the second multi-wire harness connector with the proprietary interface port, and connecting the power signal provider with a power source.

[0015] Unplugging the vehicle connector cable from the proprietary interface port may comprise unplugging a first multi-wire vehicle cable connector of the vehicle connector cable from a proprietary interface port multi-wire connector of the proprietary interface port.

[0016] Unplugging the vehicle connector cable from the vehicle module may comprise unplugging a second multi-wire vehicle cable connector of the vehicle connector cable from a vehicle module multi-wire connector of the vehicle module.

[0017] Connecting the first multi-wire harness connector with the vehicle module may comprise connecting the first multi-wire harness connector with a vehicle module multi-wire connector of the vehicle module.

[0018] Connecting the second multi-wire harness connector with the proprietary interface port may comprise connecting the second multi-wire harness connector with a proprietary interface port multi-wire connector of the proprietary interface port.

[0019] The power signal provider may comprise a fuse tap and connecting the power signal provider with the power source may comprise inserting the fuse tap into a slot of a fuse box.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] Exemplary non-limiting embodiments of the present invention are described with reference to the accompanying drawings in which:

[0021] FIG. 1 is a schematic diagram of an exemplary telematics system including a plurality of telematics devices coupled to a plurality of vehicular assets;

[0022] FIG. 2 is a block diagram showing a telematics device coupled to an asset's communications bus via an interface port;

[0023] FIG. 3 depicts the pinout of a J1962 port connector, also known as an On Board Diagnostics (OBD) port connector;

[0024] FIG. 4 depicts a telematics device and a diagnostic tool both connected to an asset's communications bus via a T-Harness;

[0025] FIG. 5 is a perspective view of an exemplary T-Harness for use with an OBD (J1962) port connector;

[0026] FIG. 6 is a simplified block diagram showing an engine control module (ECM) connected to a proprietary interface port;

[0027] FIG. 7 is a simplified block diagram of a T-harness suitable for connecting to the proprietary interface port of FIG. 6;

[0028] FIG. 8A is a block diagram showing an ECM, a proprietary interface port, and a vehicle connector cable therebetween;

[0029] FIG. 8B is a block diagram of the ECM, the proprietary interface port, and the vehicle connector cable of FIG. 8A with the vehicle connector cable disconnected from the ECM and the proprietary interface port;

[0030] FIG. 9 depicts a receptacle connector, a plug connector, male sockets, and female sockets;

[0031] FIG. 10 depicts a T-harness for installation between an ECM and a proprietary interface port, in accordance with embodiments of the present disclosure;

[0032] FIG. 11 is a perspective view depicting the insertion of a telematics device OBD interface connector into an OBD port connector end of the T-harness of FIG. 10;

[0033] FIG. 12 is a front view schematic of a splitter harness for connecting a vehicle module with a proprietary interface port, in accordance with embodiments of the present disclosure;

[0034] FIG. 13 is a side view schematic of the splitter harness of FIG. 12;

[0035] FIG. 14 is a front view schematic of a main harness segment, splitter connector, and a portion of the OBD port harness segment of the splitter harness of FIG. 12;

[0036] FIG. 15 is a front view schematic of a portion of the splitter harness of FIG. 12 including the OBD port harness segment and a power signal provider in the form of a fuse tap and power signal line;

[0037] FIG. 16 is a front view schematic showing a portion of an OBD port harness segment of the splitter harness of FIG. 12 including an OBD port connector and the power signal line providing a power signal to the OBD port connector; and

[0038] FIG. 17 is a flowchart depicting a method of installing the splitter harness of FIG. 12, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

Telematics System

[0039] FIG. 1 shows a high-level block diagram of a telematics system **101**. The telematics system **101** includes a telematics server **130**, (N) telematics devices shown as telematics device **200_1**, telematics device **200_2** . . . through telematics device **200_N** (“telematics device **200**”), a network **50**, administration terminal **140**, and operator terminals **150_1**, **150_2** . . . through **150_N** (“the operator terminals **150**”). FIG. 1 also shows a plurality of (N) assets named as asset **100_1**, asset **100_2** . . . asset **100_N** (“asset **100**”) coupled to the telematics device **200_1**, telematics device **200_2** . . . telematics device **200_N**, respectively. Additionally, FIG. 1 shows a plurality of satellites **170_1**, **170_2** and **170_3** (“the satellites **170**”) in communication with the telematics devices **200** for facilitating navigation.

[0040] The assets **100** shown are in the form of vehicles. For example, the asset **100_1** is shown as a truck, which may be part of a fleet that delivers goods or provides services. The asset **100_2** is shown as a passenger car. The asset **100_3** is shown as an electric vehicle (EV). Other types of vehicles, which are not shown, are also contemplated in the various embodiments of the present disclosure, including but not limited to, farming vehicles, construction vehicles, military vehicles, and the like.

[0041] While the assets shown in FIG. 1 are all land vehicles, this is not always the case. In some implementations, the asset is a marine vehicle or an airborne vehicle employing an ICE, an electric motor, or any other engine such as a jet engine, a rocket propulsion engine, and so on. In some cases, the asset is a piece of equipment such as a generator, a concrete mixer, a compressor, and the like.

[0042] The telematics devices **200** are coupled to assets **100**. For example, in FIG. 1 the telematics

device **200_1** is coupled to the asset **100_1**. Similarly, the telematics device **200_2** is coupled to the asset **100_2** and the telematics device **200_3** is coupled to the asset **100_3**.

[0043] The network **50** may be a single network or a combination of networks such as a data cellular network, the Internet, and other network technologies. The network **50** provides connectivity between the different components of the system, such as between the telematics devices **200** and the telematics server **130**, between the administration terminal **140** and the telematics server **130**, and between the operator terminals **150** and the telematics server **130**.

[0044] In some implementations of the telematics system **101**, the network **50** is a cellular network utilizing cellular technology. In some implementations of the telematics system **101**, the network **50** comprises a Wide Area Network (WAN) using non-cellular WAN technologies. In some implementations, the network **50** is a combination of cellular and non-cellular technologies.

[0045] The telematics server **130** is an electronic device executing machine-executable programming instructions which enable the telematics server **130** to store and analyze telematics data. The telematics server **130** may be a single computer system or a cluster of computers.

[0046] The satellites **170** are part of a global navigation satellite system (GNSS) which is a satellite-based navigation system that provides positioning, navigation, and timing services worldwide. The location information may be processed by a location module on the telematics device **200** to provide location data indicating the location of the telematics device **200** (and hence the location of the asset **100** coupled thereto).

[0047] The administration terminal **140** is an electronic device capable of connecting to the telematics server **130**, over the network **50**. The administration terminal **140** can be configured to retrieve data and analytics related to one or more assets **100**; to receive alerts from the telematics server **130** in respect of one or more conditions on the telematics device **200**; and/or to issue commands to one or more telematics device **200** via the telematics server **130**. In some implementations, an administrator **11** communicates with the telematics server **130** using the administration terminal **140**. In addition to retrieving data and analytics, the administration terminal **140** allows the administrator **11** to set alerts and geofences for keeping track of the assets **100**, receiving notifications of deliveries, receiving notifications of vehicle conditions, and receiving alerts pertaining to driver behavior.

[0048] The operator terminals **150** are electronic devices, similar to the administration terminals **140**. The operator terminals **150** are shown as smartphones, however, this is not necessarily the case. An administration terminal is any one of: a desktop computer, an industrial human-machine interface (HMI), a touch screen panel, a table, a smartphone, an Augmented Reality (AR) headset, and a Network Operations Center (NOC). The operator terminals **150** are used by operators **10** (for example, vehicle drivers) of the assets **100** to both track and configure the usage of the assets **100**. For example, as shown in FIG. 1, the operator **10_1** has the operator terminal **150_1**, the operator **10_2** has the operator terminal **150_2**, and the operator **10_N** has the operator terminal **150_N**. Assuming the operators **10** all belong to a fleet of vehicles, each of the operators **10** may operate any of the assets **100**. For example, FIG. 1 shows that the operator **10_1** is associated with the asset **100_1**, the operator **10_2** is associated with the asset **100_2**, and the operator **10_N** is associated with the asset **100_N**. However, any operator **10** may operate any asset **100** within a particular group of assets, such as a fleet. The operator terminals **150** are in communication with the telematics server **130** over the network **50**.

[0049] In operation, a telematics device **200** is coupled to an asset **100** to capture asset data. In some implementations, the asset data is combined with location data obtained by the telematics device **200** from a location module in communication with the satellites **170** and/or sensor data gathered from sensors in the telematics device **200** or another device coupled to the telematics device **200**. The combined asset data, location data, and sensor data are termed “telematics data.” The telematics device **200** sends the telematics data **212** to the telematics server **130** over the network **50**. The telematics server **130** processes, aggregates, and/or analyzes the telematics data

212 to generate asset information pertaining to the assets **100** or to a fleet of assets. In some implementations, the telematics server **130** stores the telematics data **212** and/or the generated asset information in the telematics database **132**. In some implementations, the administration terminal **140** connects to the telematics server **130**, over the network **50**, to access the generated asset information. In other implementations, the telematics server **130** pushes the generated asset information to the administration terminal **140**. In some implementations, the operators **10** use the operator terminals **150** to indicate to the telematics server **130** which assets **100** they are associated with. In response, the telematics server **130** updates the telematics database **132** to associate an operator **10** with an asset **100**. In some implementations, the telematics server **130** provides additional analytics related to the operators **10** including work time, location, and operating parameters. For example, for vehicle assets, the telematics data **212** may include turning, speeding, and braking information. The telematics server **130** can correlate the telematics data to the vehicle's driver by querying the telematics database **132** for a particular vehicle and retrieving the associated driver information. In some implementations, an administrator **11** uses the administration terminal **140** to set alerts for certain activities pertaining to the assets **100**. When criteria for an alert is met, the telematics server **130** sends a message to the administration terminal **140** to notify an administrator **11**. In some implementations, the telematics server **130** sends alerts to the operator terminal **150** to notify an operator **10** of the alert. In some implementations, the telematics server **130** sends alerts to the telematics device **200** to generate an alert to the driver such as a beep, a displayed message, or an audio message.

[0050] The asset **100** may have a plurality of electronic control units (ECUs) of the above-mentioned types. A vehicle may, for example, have around seventy ECUs. For simplicity, only a few of the ECUs **110** are depicted in FIG. 2. For example, in the depicted embodiment the asset **100** has three ECUs shown as the ECU **110A**, the ECU **110B**, and the ECU **110C** (“the ECUs **110**”). The ECU **110A**, the ECU **110B**, and the ECU **110C** are shown to be interconnected via an asset communications bus.

[0051] The most commonly used type of asset communications bus is the Controller Area Network (CAN) bus. CAN is a robust and standardized communication protocol designed for real-time control applications. The CAN bus is a physical bus used to connect various ECUs and sensors, allowing them to exchange data and commands. CAN ensures that different vehicle systems can work together seamlessly. ECUs are connected to the CAN bus using dedicated CAN transceivers and connectors.

[0052] For example, in FIG. 2 the ECUs **110** are interconnected using the CAN bus **104**. The ECUs **110** send and receive information to one another in CAN data frames by placing the information on the CAN bus **104**. When an ECU **110** places information on the CAN bus **104**, other ECUs **110** receive the information and may or may not consume or use that information.

[0053] Different protocols may be used to exchange information between the ECUs over a CAN bus. For example, ECUs **110** in trucks and heavy vehicles use the Society of Automotive Engineering (SAE) J1939 protocol to exchange information over a CAN bus **104**. J1939 is based on CAN and is used for diagnostic and communication purposes. Most passenger vehicles use the SAE J1979 protocol, which is commonly known as On-Board Diagnostic II (OBD-II) protocol to exchange information between ECUs **110** on their CAN bus **104**. OBD-II is a standardized diagnostic protocol used in most vehicles manufactured since the late 1990s. OBD-II provides a common interface for diagnostic tools to communicate with a wide range of ECUs in the vehicle, including the engine control module (ECM), transmission control module (TCM), and more. OBD-II allows for reading diagnostic trouble codes (DTCs), live data, and performing various diagnostic tests.

[0054] An asset **100** may allow access to information exchanged over the CAN bus **104** via an interface port **102**. For example, if the asset **100** is a passenger car, then the interface port **102** is most likely an OBD-II port (“OBD port”). Data accessible through the interface port **102** is termed

the asset data **112**. In some implementations, the interface port **102** includes a power interface for providing electric power to a telematics device **200** connected thereto.

Telematics Device

[0055] Further details relating to the telematics device **200** and how it interfaces with an asset **100** are shown with reference to FIG. 2. FIG. 2 depicts an asset **100** and a telematics device **200** coupled thereto. Selected relevant components of each of the asset **100** and the telematics device **200** are shown.

[0056] The telematics device **200** includes a controller **230** coupled to a memory **240**, an asset interface **202** and a network interface **220**. The telematics device **200** also includes one or more sensors **204** and a location module **206**. In some implementations, the telematics device **200** contains an inertial measurement unit, shown as the IMU **290**. The telematics device **200** may also contain some optional components, shown in dashed lines in FIG. 2. For example, the telematics device **200** may contain one or more of: a near-field communications (NFC) module such as NFC module **260**, a short-range wireless communications module **270**, and a wired communications module such as a serial communications module **280**. In some embodiments (not shown), the telematics device **200** may have a dedicated power source or a battery. In other embodiments, the telematics device **200** may receive power directly from the asset **100**, via the interface port **102**. The telematics device **200** shown is an example. Some of the components shown in solid lines may also be optional and may be implemented in separate modules. For example, some telematics devices (not shown) may not have a location module **206** and may rely on an external location module for obtaining the location data. Some telematics devices may not have any sensors **204** and may rely on external sensors for obtaining sensor data.

Interface Port

[0057] FIG. 3 depicts the pinout of a vehicle J1962 16-pin connector, also referred to as an OBD port connector **300**. As shown in FIG. 3, the OBD port connector **300** has different signal lines which may be used by different protocols. Signal line 1 is known as the SAE J1850 bus+ (“J1850+”) signal line and signal line 2 is known as the SAE J1850 bus- (“J1850-”) signal. The J1850+/J1850- signal lines are used by the Variable Pulse Width (VPW) protocol and the Pulse Width Modulation (PWM) protocol. Signal line 7 is known as the ISO9141 K-line (“K-line”) signal line while signal line 15 is known as the ISO9141 L-line (“L-line”) signal line. The K-line and L-line are used by the ISO9141 protocol and the ISO14230 protocol. Signal line 6 is the ISO 15765-4 CAN High (“CANH”) signal line while signal line 14 is the ISO 15765-4 CAN Low (“CANL”) signal line. The CANH and CANL signal lines are used by the CAN protocol.

[0058] In addition to having a telematics device **200** connected therewith, it may also be desired to connect a diagnostic tool such as an OBD reader or an emissions testing device to the vehicle asset. Since a vehicle may only have one interface port (e.g., OBD port), a splitting device needs to be used. FIG. 4 depicts the asset **100** of FIG. 2 shown connected to both a telematics device **200** and a diagnostic tool **600** via a splitter harness. As an example of a splitter harness, a T-harness **400** that splits a group of signals into two groups of signals is shown.

[0059] In this disclosure a “diagnostic tool” is an electronic device that may be used to read asset data **112** for the purpose of diagnosing problems, for the purpose of conducting performance testing such as emission testing, or for the purpose of ECU programming and/or firmware update. A diagnostic tool may also clear some engine error codes, typically known as Diagnostic Trouble Codes (DTCs). A diagnostic tool may also be referred to as a “scan tool” or a “testing tool”.

[0060] A wire harness is a specially designed system that keeps numerous wires organized. In this disclosure, a “splitter harness” refers to a harness device that splits a group of signals into multiple groups of signals. As a non-limiting embodiment, a “T-harness” splits a group of signals into two groups of signals thus allowing two electrical devices to connect to the same interface. In the context of a vehicle a splitter harness can be used to connect multiple devices to the interface port **102** of a vehicle. A T-harness can be used to connect two devices to the interface port (e.g. the OBD

port) of a vehicle. Logically, a splitter harness consists of a splitter connector and a plurality of cables or harness segments each ending in a connector for interfacing to a device. The splitter connector depicted in FIG. 4 comprises a T-connector **450**, a first harness segment **410**, a second harness segment **420**, and a third harness segment **440**. The T-connector **450** splits the group of signals provided by the interface port **102** over the first harness segment **410**, such as the signals of the OBD port connector **300**, into two groups of signals each sent over the second harness segment **420** and the third harness segment **440**. For example, a telematics device **200** may connect to the second harness segment **420** and a diagnostic tool **600** may connect to the third harness segment **440**. Accordingly, both the telematics device **200** and the diagnostic tool **600** have access to the interface port **102** and the CAN bus **104**.

[0061] FIG. 5 depicts a T-harness **500** for connecting to a vehicle with an OBD port and having a standard OBD (J1962) 16-wire connector. The T-harness **500** has a vehicle OBD harness connector **510**, a telematics device harness segment **520**, a telematics device OBD port connector **530**, an auxiliary OBD harness segment **540**, and an auxiliary OBD harness connector **550**.

[0062] The vehicle OBD harness connector **510** connects the T-harness **500** to the vehicle's own OBD connector. The vehicle OBD harness connector **510** also serves as a splitter connector (in this case a T-connector) which splits the group of signals of the vehicle OBD harness connector **510** into two identical groups of signals. A first group of signals goes into the telematics device harness segment **520**, and a second group of signals goes into the auxiliary OBD harness segment **540**. In other words, every signal of the OBD port is replicated in each of the telematics device harness segment **520** and the auxiliary OBD harness segment **540**.

[0063] The telematics device OBD port connector **530** is shaped for receiving a device with a mating OBD connector such as a telematics device **200**. The telematics device OBD port connector **530** is connected to the telematics device harness segment **520**.

[0064] The auxiliary OBD harness connector **550** is intended to replace the vehicle's OBD connector. The auxiliary OBD harness connector **550** is connected to the auxiliary OBD harness segment **540**.

[0065] In a typical arrangement, the T-harness **500** is installed in a vehicle to replace the vehicle's OBD port and yet provide an additional OBD port for connecting a telematics device thereto. To install the T-harness **500** in a vehicle, the vehicle's own OBD port is removed from the vehicle mount, and connected to the vehicle OBD harness connector **510**. The auxiliary OBD harness connector **550** is connected to the vehicle mount to replace the vehicle's OBD connector. The auxiliary OBD harness connector **550** may need an adapter to allow it to fit to the vehicle mount. By connecting the auxiliary OBD harness connector **550** to the vehicle mount, the vehicle complies with some regulations such as the California Air Resources Board (CARB) which mandates that the location of a vehicle's OBD port is not changed. The auxiliary OBD harness connector **550** then allows any OBD devices to connect to the vehicle's asset communications bus for example to conduct emissions testing.

[0066] Finally, a telematics device **200** may connect to the telematics device OBD port connector **530**. It should be noted that any other device that has a standard OBD connector J1962 may connect to the telematics device OBD port connector **530**. Advantageously, the vehicle still provides an OBD port at the same location as the original OBD port thereof, while at the same time also allowing a telematics device **200** to connect to the asset communications bus and capture telematics data.

Proprietary Interface Ports

[0067] Vehicles manufactured prior to **1996** may have proprietary interface ports having a proprietary interface port connector, instead of the standard **J1962** OBD connector. Some of such vehicles do not have many instances of the ECU **110**, connected by a CAN bus. Instead many such vehicles have a vehicle module such as an engine control module (ECM) connected with the proprietary interface port. Such an arrangement is depicted in FIG. 6. In FIG. 6, the ECM **160** is

connected to a proprietary interface port **620**.

[0068] Modern telematics devices are designed to fit into an OBD port connector based on the J1962 standard, such as the OBD port connector **300** of FIG. 3. A telematics device **200** having an OBD interface connector configured to mate with the OBD port connector **300** cannot connect directly to a proprietary interface port connector having a different shape from the OBD port connector **300**. Furthermore, if it is desired to connect a telematics device to an older vehicle (pre-1996) while having the proprietary interface port available for other devices to connect thereto, then a proprietary T-harness needs to be made. The proprietary T-harness can facilitate making the proprietary interface port available (e.g., for connectivity with a proprietary diagnostic tool) while also allowing the connection of a telematics device that has an OBD interface connector. A simplified block diagram of a proprietary T-harness for connecting to a proprietary interface port is shown in FIG. 7. The proprietary T-harness **700** contains a proprietary harness connector **710**, a T-connector **715**, an auxiliary proprietary harness connector **750**, and an OBD port connector **730**. The various connectors are connected by harness segments as shown.

[0069] The proprietary harness connector **710** is a proprietary connector having the opposite gender of the proprietary interface port connector and is therefore designed to mate therewith. The T-connector **715** splits the group of signals from the proprietary harness connector **710** into two groups of signals. In other implementations, the T-connector **715** is not a separate component. Instead, the proprietary harness connector **710** may contain a T-connector for splitting the signals of the proprietary interface port **620** into two groups of signals for routing to the auxiliary proprietary harness connector **750**, and the OBD port connector **730**. The auxiliary proprietary harness connector **750** is similar to the proprietary interface port connector. The OBD port connector **730** is similar to the aforementioned telematics device OBD port connector **530**.

[0070] While the proprietary T-harness **700** would work, the proprietary T-harness **700** requires tooling two proprietary connectors of opposite genders, namely the proprietary harness connector **710** and the auxiliary proprietary harness connector **750**. The tooling of two proprietary connectors is costly and time consuming.

[0071] Another limitation of the proprietary T-harness **700** is that it may not provide power to the OBD port connector **730**, which is a requirement for OBD ports. For example, a telematics device connected to the OBD port connector **730** will not work if the proprietary interface port **620** does not provide any power signals to the proprietary T-harness **700** via the proprietary harness connector **710**.

[0072] In some vehicles, a vehicle module such as the ECM **160** is connected with the proprietary interface port **620** by a vehicle connector cable **800**. FIG. 8A and FIG. 8B show such an arrangement. A vehicle module can have a vehicle module multi-wire connector for connecting the vehicle module to the proprietary interface port, for example by a cable. For example, the ECM **160** has a vehicle module multi-wire connector in the form of a male plug **840**. The proprietary interface port **620** has a proprietary interface port connector **622** for connecting devices to the vehicle, and a proprietary multi-wire connector in the form of a female receptacle **850** for connecting the proprietary interface port. A vehicle connector cable has a first multi-wire vehicle cable connector at a first end, a second multi-wire vehicle cable connector at a second end, and conduit in which signal lines run between the first multi-wire connector and the second multi-wire connector. As an example, the vehicle connector cable **800** has a conduit **810** in which the signal lines are running between a male plug **830** at one end and a female receptacle **820** at the other end. The male plug **830** is configured to mate with the female receptacle **850** of the proprietary interface port **620**. Similarly, the male plug **840** is configured to mate with the female receptacle **820** of the ECM **160**. When the male plug **830** is mated with the female receptacle **850**, and the male plug **840** is mated with the female receptacle **820**, electrical connectivity is established between the ECM **160** and the proprietary interface port **620**.

[0073] The male plug **830**, the male plug **840**, the female receptacle **820**, and the female receptacle

850 are off-the-shelf connectors such as the Molex™ Micro-Fit 3.0 series shown in FIG. **9**. A 6-pin male plug **920** can receive 6 female sockets **925**. A 6-pin female receptacle **930** receives 6 male sockets **935**. The 6-pin male plug **920** can mate with the 6-pin female receptacle **930** to establish electrical connectivity.

[0074] The inventors have come up with a splitter harness for providing connectivity between a vehicle module and a proprietary interface port. The vehicle module has a vehicle module multi-wire connector and the proprietary interface port has a proprietary interface port multi-wire connector. The splitter harness is comprised of a main harness segment, an OBD port harness segment, a signal splitter, and a power signal provider. The main harness segment has a first multi-wire harness connector at a first end thereof and a second multi-wire harness connector at a second end thereof. The OBD port harness segment has an OBD port connector at a first end thereof. The signal splitter is a device that splits a group of signal lines of the main harness segment into a first group of signal lines routed on the main harness segment to the second multi-wire harness connector, and a second group of signal lines routed to a second end of the OBD port harness segment. The splitter harness connects a proprietary interface port to a vehicle module and provides an OBD port connector for connecting a device such as a telematics device. The power signal provider provides a power signal to the OBD port connector. As an example, a T-harness connecting an ECM to a proprietary interface port will be discussed below. In this implementation, the ECM represents a vehicle module.

[0075] FIG. **10** depicts an exemplary implementation of a splitter harness in the form of a T-harness **1000**, shown FIG. **10**. The T-harness **1000** can provide connectivity between the ECM **160** and the proprietary interface port **620**, provide OBD signals at an OBD port connector **1530**, and provide power to the OBD port connector **1530**.

[0076] The T-harness **1000** has a main harness segment **1010** having a first multi-wire harness connector in the form of a male plug **1030** at one end thereof and a second multi-wire harness connector in the form of a female receptacle **1020** at the opposite end thereof. The male plug **1030** and the female receptacle **1020** are off-the-shelf components similar to the ones depicted in FIG. **9**. The main harness segment **1010** also has a signal splitter **1015**. The OBD port harness segment **1060** is connected at one end thereof to the signal splitter **1015** and is connected at another end thereof to an OBD port connector **1530**. A power signal line **1012** connects a fuse tap **1025** to the OBD port connector **1530**.

[0077] The main harness segment **1010** contains a group of signal lines running between the male plug **1030** and the female receptacle **1020**.

[0078] The male plug **1030** and the female receptacle **1020** are multi-wire harness connectors similar to the 6-pin male plug **920** and the 6-pin female receptacle **930** of FIG. **9**, respectively. The male plug **1030** and the female receptacle **1020** may, however, have any number of pins based on the number of signal lines between an engine control module and a proprietary interface port.

[0079] The signal splitter **1015** is a device that splits the group of signal lines from the female receptacle **1020** into two groups of signals, routes a first group of signals on the main harness segment to the male plug **1030**, and routes a second group of signal lines to the OBD port harness segment **1060**. The signal splitter **1015** may be a T-connector or a Y-connector.

[0080] The OBD port harness segment **1060** routes the second group of signal lines from the main harness segment **1010** to the OBD port connector **1530**. The OBD port connector **1530** is similar to the telematics device OBD port connector **530**.

[0081] The power signal provider of the T-harness **1000** is in the form of a fuse tap **1025**. The fuse tap **1025** is an off-the-shelf component that connects to a fuse circuit in a fuse box **145** of a vehicle, and provides a power signal that is transmitted over the power signal line **1012** to the OBD port harness segment **1060**. The fuse tap **1025** typically has two slots for connecting two fuses; one fuse controls the power signal that is transmitted over the power signal line **1012** and another fuse for controlling the original circuit that the fuse circuit represents (if applicable)

[0082] FIGS. **12-16** are engineering drawings of the T-harness **1000**, in accordance with embodiments of the present disclosure.

[0083] FIG. **12** depicts the T-harness **1000** in a front view thereof.

[0084] FIG. **13** is a side view taken from the side of the OBD port connector **1530**. FIG. **13** shows a fuse **1026** connected to the fuse tap **1025** for enabling the power signal to be routed to the OBD port connector **1530**.

[0085] FIG. **14** is a front view focusing on the main harness segment **1010**, the male plug **1030** and the female receptacle **1020**. The main harness segment **1010** is comprised of a plurality of signal lines bundled together by cable ties **1019**. The main harness segment **1010** has a T connection from which signal lines are directed to become the OBD port harness segment **1060**. The signal lines are routed inside a conduit having a heat shrink **1062** at ends thereof. A label **1064** may optionally be connected to the conduit and include relevant information about the T-harness **1000**.

[0086] FIG. **15** is a front view focusing on a central portion of the T-harness **1000**. A portion of the OBD port harness segment **1060** is shown. Furthermore, the fuse tap **1025** is shown in more detail. The fuse tap **1025** has two leads **1027** for insertion into a fuse slot in a fuse box. The fuse tap **1025** provides a power signal line **1012** and two fuse slots. In FIG. **15** one fuse slot contains a fuse **1026** which connects power from the leads **1027** to the power signal line **1012**. The second fuse slot **1028** is available. If the fuse slot of the fuse box had an original fuse, that original fuse may be inserted in the second fuse slot **1028**.

[0087] FIG. **16** is a front view focusing on the OBD port connector end of the T-harness **1000** and shows the signal lines from the OBD port harness segment **1060** connected to the OBD port connector **1530**. FIG. **16** also shows the conduit and the heat shrink **1062**. The power signal line **1012** from the fuse tap **1025** is also shown connected to the OBD port connector **1530** for providing power to pin 16 (as shown in FIG. **3**, pin 16 is the power pin).

[0088] To install the T-harness **1000**, first the vehicle connector cable **800** is disconnected at both ends from the ECM **160** and the proprietary interface port **620**. Specifically, as shown in FIG. **8**, the male plug **830** is unplugged from the female receptacle **850** and the male plug **840** is unplugged from the female receptacle **820**.

[0089] Next, the female receptacle **1020** of the T-harness **1000** is connected to the male plug **840** of the ECM **160**.

[0090] Next, the male plug **1030** of the T-harness **1000** is plugged into the female receptacle **850** of the proprietary interface port **620**.

[0091] Next, the fuse tap **1025** is inserted in an available fuse slot in the fuse box **145**. Typically a fuse tap **1025** allows for the connection of another fuse thereto. Accordingly a fuse of the appropriate rating may be removed from the slot thereof and inserted into the fuse tap **1025**. Additionally, a fuse **1026** is inserted into the fuse tap for routing the power signal to the power signal line, then the fuse tap **1025** is inserted into the use slot in the fuse box **145**.

[0092] Lastly, a telematics device or any diagnostic device that has an OBD connector may be plugged into the OBD port connector **1530**, as shown in FIG. **11**.

[0093] Advantageously, the provided T-harness **1000** allows the connection of a diagnostic tool or a telematics device having an OBD connector, to a vehicle that has a proprietary interface port **620**. Additionally, the T-harness provides power to the diagnostic tool or telematics device if needed by using a fuse tap that is part of the T-harness **1000**. The provided T-harness **1000** only contains off-the-shelf components and is therefore inexpensive to manufacture. Another advantage of using the T-harness **1000** is that on older vehicles the pins of the proprietary interface port connector **622** may have poor conductivity that impacts signal quality. Connecting a telematics device to the pins of the proprietary interface port connector **622** may lead to poor capturing of the vehicle data by the telematics device **200**. The T-harness **1000**, however, connects to the ECM directly and is therefore unaffected by the age and/or condition of the proprietary interface port connector **622**.

[0094] While the T-harness **1000** uses a female receptacle **1020** at the end which connects to the

ECM **160** and a male plug **1030** at the end which connects to the proprietary interface port **620**, this is not necessarily the case. In some embodiments, the situation may be reversed, both ends may have male plugs, or both ends may have female receptacles. This will all depend on the type of multi-wire connectors used by the ECM **160** and the proprietary interface port **620**.

[0095] The proprietary interface port **620** is shown to have proprietary interface port multi-wire connector (in the form of the female receptacle **850**) directly connected to the proprietary interface port connector **622**. However, this is not necessarily the case, and the female receptacle **850** may connect to the proprietary interface port connector **622** via a cable.

[0096] The signal splitter **1015** is shown located on the main harness segment **1010** halfway between the male plug **1030** and the female receptacle **1020**. However, it is also possible that a splitting female receptacle is used instead of the female receptacle **1020**, the splitting female receptacle splitting the signals into the main harness segment **1010** and the OBD port harness segment **1060**.

[0097] The T-harness **1000** may obtain power from other sources in the vehicle other than the fuse box, however a fuse tap **1025** is a convenient and clean method that does not involve splicing wires.

[0098] A method of installing a splitter harness, such as the T-harness **1000** in a vehicle having a proprietary interface port connected to a vehicle module via a vehicle connector cable is shown in FIG. **17**. The splitter harness comprises a main harness segment having a first multi-wire harness connector and a second multi-wire harness connector at both ends thereof. The splitter harness further comprises an OBD port harness segment having an OBD port connector at an end thereof, a splitter signal for splitting a group of signal lines on the main harness segment into a first group of signal lines routed to the second multi-wire harness connector and a second group of signal lines routed to the OBD port harness segment. The splitter harness also comprises a power signal provider for providing power to the OBD port connector. The method starts at step **1710**.

[0099] At step **1710**, the vehicle connector cable is unplugged from the proprietary interface port. For example, with reference to FIG. **8B**, this step involves unplugging the male plug **830** from the female receptacle **850** of the proprietary interface port **620**.

[0100] At step **1720**, the vehicle connector cable is unplugged from the vehicle module. For example, with reference to FIG. **8B**, this step involves unplugging the female receptacle **820** from the male plug **840**.

[0101] At step **1730**, the first multi-wire harness connector is connected with the vehicle module. For example, with reference to FIG. **10**, this step involves connecting the female receptacle **1020** with the male plug **840** of the ECM **160**.

[0102] At step **1740**, the second multi-wire harness connector is connected with the proprietary interface port. For example, with reference to FIG. **10**, this involves connecting the male plug **1030** with the female receptacle **850** of the proprietary interface port **620**.

[0103] At step **1750**, the power signal provider is connected with a power source. For example, with reference to FIG. **10**, this involves connecting the fuse tap **1025** to a power source such as the fuse box **145**.

[0104] Embodiments have been described where the techniques are implemented in circuitry and/or computer-executable instructions. It should be appreciated that some embodiments may be in the form of a method or process, of which at least one example has been provided. The acts performed as part of the method or process may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments. Various aspects of the embodiments described above may be used alone, in combination, or in a variety of arrangements not specifically discussed in the embodiments described in the foregoing and is therefore not limited in its application to the details and arrangement of components set forth in the foregoing description or illustrated in the drawings. For

example, aspects described in one embodiment may be combined in any manner with aspects described in other embodiments.

Claims

- 1.** A splitter harness for use in a vehicle having a proprietary interface port and a vehicle module, the splitter harness comprising: a main harness segment having: a first multi-wire harness connector at a first end thereof, and a second multi-wire harness connector at a second end thereof; an On-Board Diagnostic (OBD) port harness segment having an OBD port connector at a first end thereof; a signal splitter for splitting a group of signal lines of the main harness segment into a first group of signal lines routed on the main harness segment to the second multi-wire harness connector and a second group of signal lines routed to a second end of the OBD port harness segment; and a power signal provider for providing a power signal from a power source to the OBD port connector.
- 2.** The splitter harness of claim 1, wherein the first multi-wire harness connector is configured for connecting to a vehicle module multi-wire connector of the vehicle module.
- 3.** The splitter harness of claim 1, wherein the second multi-wire harness connector is configured for connecting to a proprietary interface port multi-wire connector of the proprietary interface port.
- 4.** The splitter harness of claim 2, wherein the first multi-wire harness connector comprises a male plug and the vehicle module multi-wire connector comprises a female receptacle.
- 5.** The splitter harness of claim 2, wherein the first multi-wire harness connector comprises a female receptacle and the vehicle module multi-wire connector comprises a male plug.
- 6.** The splitter harness of claim 3, wherein the second multi-wire harness connector comprises a male plug and the proprietary interface port multi-wire connector comprises a female receptacle.
- 7.** The splitter harness of claim 3, wherein the second multi-wire harness connector comprises a female receptacle and the proprietary interface port multi-wire connector comprises a male plug.
- 8.** The splitter harness of claim 1, wherein the power signal provider comprises a power signal line and a fuse tap.
- 9.** The splitter harness of claim 1, wherein the signal splitter comprises a splitter connector located on the main harness segment.
- 10.** The splitter harness of claim 1, wherein the signal splitter comprises a splitter connector which is part of the first multi-wire harness connector.
- 11.** A method of installing a splitter harness in a vehicle having a proprietary interface port connected to a vehicle module via a vehicle connector cable, the splitter harness comprising a main harness segment having a first multi-wire harness connector and a second multi-wire harness connector at both ends thereof, an On-Board Diagnostic (OBD) port harness segment having an OBD port connector at an end thereof, a signal splitter for splitting a group of signal lines on the main harness segment into a first group of signal line routed to the second multi-wire harness connector and a second group of signal lines routed to the OBD port harness segment, and a power signal provider for providing power to the OBD port connector, the method comprising: unplugging the vehicle connector cable from the proprietary interface port; unplugging the vehicle connector cable from the vehicle module; connecting the first multi-wire harness connector with the vehicle module; connecting the second multi-wire harness connector with the proprietary interface port; and connecting the power signal provider with a power source.
- 12.** The method of claim 18, wherein unplugging the vehicle connector cable from the proprietary interface port comprises unplugging a first multi-wire vehicle cable connector of the vehicle connector cable from a proprietary interface port multi-wire connector of the proprietary interface port.
- 13.** The method of claim 18, wherein unplugging the vehicle connector cable from the vehicle module comprises unplugging a second multi-wire vehicle cable connector of the vehicle connector

cable from a vehicle module multi-wire connector of the vehicle module.

14. The method of claim **18**, wherein connecting the first multi-wire harness connector with the vehicle module comprises connecting the first multi-wire harness connector with a vehicle module multi-wire connector of the vehicle module.

15. The method of claim **18**, wherein connecting the second multi-wire harness connector with the proprietary interface port comprises connecting the second multi-wire harness connector with a proprietary interface port multi-wire connector of the proprietary interface port.

16. The method of claim **18**, wherein: the power signal provider comprises a fuse tap; and connecting the power signal provider with the power source comprises inserting the fuse tap into a slot of a fuse box.

17. The method of claim **18**, further comprising plugging a telematics device to the OBD port connector.

18. A method of installing the splitter harness of claim 1, comprising: unplugging a vehicle connector cable from the proprietary interface port; unplugging the vehicle connector cable from the vehicle module; connecting the first multi-wire harness connector with the vehicle module; connecting the second multi-wire harness connector with the proprietary interface port; and connecting the power signal provider with a power source.

19. The method of claim 18, wherein unplugging the vehicle connector cable from the proprietary interface port comprises unplugging a first multi-wire vehicle cable connector of the vehicle connector cable from a proprietary interface port multi-wire connector of the proprietary interface port.

20. The method of claim 18, wherein unplugging the vehicle connector cable from the vehicle module comprises unplugging a second multi-wire vehicle cable connector of the vehicle connector cable from a vehicle module multi-wire connector of the vehicle module.

21. The method of claim 18, wherein connecting the first multi-wire harness connector with the vehicle module comprises connecting the first multi-wire harness connector with a vehicle module multi-wire connector of the vehicle module.

22. The method of claim 18, wherein connecting the second multi-wire harness connector with the proprietary interface port comprises connecting the second multi-wire harness connector with a proprietary interface port multi-wire connector of the proprietary interface port.

23. The method of claim 18, wherein the power signal provider comprises a power signal line and a fuse tap and wherein connecting the power signal provider with the power source comprises inserting the fuse tap into a slot of a fuse box.

24. The method of claim 18, further comprising plugging a telematics device to the OBD port connector.
